

Aritra Chatterjee

Data Engineer & System Design Architect

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Summary

A decade of experience in the domain of Data Engineering, Distributed systems, Databases, Microservices(API Gateway,Event-Driven & Containerisation) and Cloud architecture(AWS & Azure).

Technical Skills

Programming Languages

Python

Bash

R

Big Data

Kafka

Spark

Orchestration & Containerization

Docker

Kubernetes

Web Frameworks

Flask

Django

Microservices

Distributed Systems

RESTful APIs

Event-Driven Architecture

Cloud & DevOps

AWS

Azure

GithubActions

Terraform

Databases

PostgreSQL

CosmosDB

MongoDB

Mysql

Professional Experience

Tech Lead at Cognizant Technology Solutions Corporations

Oct-2023 – Present

- Played a key role in a cloud modernization project, transitioning messaging systems from Azure Service Bus to Confluent Kafka for enhanced performance and real-time capabilities.
- Managed data orchestration schedules using Control-M, controlling data pipeline execution and job dependencies for mission-critical Databricks (PySpark) workloads.
- Contributed to infrastructure as code (IaC) development with Terraform, provisioning and managing Azure resources including Cosmos DB for database needs and Azure Key Vault for secrets management.
- Exposure in KEDA autoscaling for Kubernetes workloads, effectively scaling pods to zero during periods of inactivity and scaling out based on message queues, reducing infrastructure costs.
- Exposure with centralized monitoring strategy with Datadog, integrating logs, metrics, and traces from Databricks, Kafka, and Kubernetes to provide comprehensive observability.

Tech Lead at Accenture

Oct-2021 – Oct-2023

- Led data platform modernization, migrating legacy COBOL and SAS systems to Python-based pipelines using Databricks and Spark.
- Architect and developed a Python-based API using AWS Lambda to automate the clinical approval process for a Merck client, reducing manual overhead.

- Increased data pipeline robustness by implementing a CI/CD workflow with Jenkins

Sr. Software Engineer at Larsen & Toubro Infotech Limited

Mar-2020 –
Oct-2021

- Architect an automated data processing framework that ingests JSON templates to dynamically apply Spark transformations on datasets.
- Generated structured JSON reports from transformed data, which directly fed predefined Tableau dashboards for business intelligence.
- Leveraged a modern tech stack, including Python, Spark, PostgreSQL, Git, AWS S3, and AWS Fargate, for a scalable and reliable solution.
- Implemented containerized deployment using Docker on AWS Fargate to ensure portability and automated scalability of the framework.

Sr. Software Engineer at Capital Numbers Ltd.

Sept-2019 –
Mar-2020

- Built an internal online code compiler supporting 14 languages using the Sphere Engine API, with Flask for URL creation and automated API documentation via Swagger.
- Designed and implemented a web-crawling system with the Scrapy framework in Python, scheduling specific domain recrawls via cron jobs, and served results through a Flask-based application.

Data Science at GSPANN Technology Ltd.

Jan-2019 –
Sep-2019

- Improved operational efficiency by 20% by building and automating data-driven dashboards in R Shiny and Python (using Dash) for executive and operational teams.

- Led end-to-end data analysis projects, from defining business problems with stakeholders to data collection, statistical modeling, and visualization.

Python & Rshiny at Online

Mar-2017 –

Freelancer

Dec-2018

- Provided analytical solutions for microeconomics and accountancy, building dynamic Excel models and dashboards for graduate-level projects. Delivered services through online portals(Chegg India), ensuring high-quality, data-driven analysis.
- Developed and deployed interactive web applications using R Shiny and Python Dash for various online projects, enhancing data visualization capabilities for clients.

Data Analytics & Modelling at

Dec-2012 –

Accenture

Feb-2017

- Built an automated survey reporting tool for Accenture by implementing an end-to-end text mining pipeline, including feature engineering and sentiment analysis for qualitative feedback.
- Developed and deployed data visualization dashboards using R Shiny and Python Dash, showcasing proficiency in both R and Python for data management and reporting.
- Facilitated the Change Management process, coordinating with stakeholders and the Change Advisory Board (CAB) to review and implement IT changes with minimal business disruption.
- Conducted root cause analysis (RCA) for recurring incidents to identify underlying problems, which led to service improvement plans and reduced repeat incident

ITSM Service Management at IBM May 2011 – **GPS** May-2012

- Utilized ITSM tools like BMC and ServiceNow to track, prioritize, and resolve incident, problem, and change tickets.
- Authored clear and concise technical documentation of bridge calls, capturing timelines, diagnostic steps, and resolution details for post-incident review and knowledge management.

Education

PGDM	2007 – 2009
Globsyn Business School	
BA in Economics(Hons), Mathematics & Statistics	2003 – 2006
North Bengal University.	